

## **Basic Information**

**Experiment No.:** 18

**Date:** 24-09-2026

**Title:** Java Queue Interface using LinkedList

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## **Objective / Aim**

To implement the Queue interface using "LinkedList" and demonstrate insertion, accessing, and removal of elements from the queue.

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## **System Requirements**

### Hardware

- Computer/Laptop
- Minimum 4 GB RAM

### Software

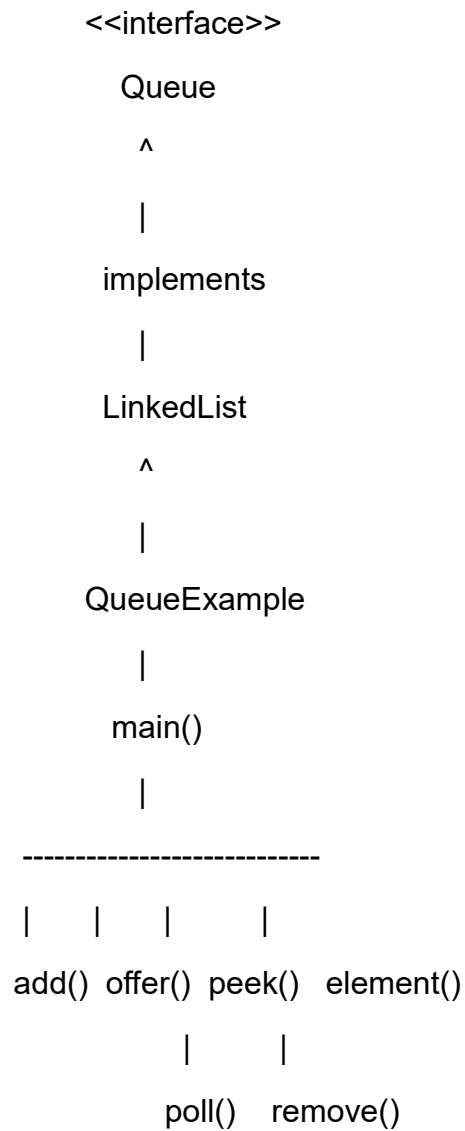
- Java JDK
  - Any Java IDE or Command Prompt
  - Operating System: Windows/Linux/macOS
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## **Algorithm / Logic Description**

1. Create a "Queue" using "LinkedList".
2. Add elements using "add()".
3. Insert an element using "offer()".
4. View the first element using "peek()" and "element()".
5. Remove the first element using "poll()".

6. Remove another element using "remove()".
  7. Display the queue after removal.
  8. End the program.
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#### UML Class Diagram



#### Source Code

```
package collectionframework;
```

```
import java.util.Queue;
import java.util.LinkedList;

public class QueueExample {
    public static void main(String[] args) {

        Queue<String> q = new LinkedList<>();

        // add()
        q.add("A");
        q.add("B");
        q.add("C");

        System.out.println("Queue: " + q);

        // offer()
        q.offer("D");
        System.out.println("After offer: " + q);

        // peek()
        System.out.println("Head using peek: " + q.peek());

        // element()
        System.out.println("Head using element: " + q.element());

        // poll()
        System.out.println("Removed using poll: " + q.poll());
```

```
// remove()

System.out.println("Removed using remove: " + q.remove());

System.out.println("Queue after removal: " + q);
}
}
```

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## **Compilation & Execution Commands**

Save the file as:

QueueExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac QueueExample.java

Run:

java collectionframework.QueueExample

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## **Output Screen**

Queue: [A, B, C]

After offer: [A, B, C, D]

Head using peek: A

Head using element: A

Removed using poll: A

Removed using remove: B

Queue after removal: [C, D]

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## **Result**

The Queue interface was successfully implemented using "LinkedList", and the "add()", "offer()", "peek()", "element()", "poll()", and "remove()" operations were successfully demonstrated.